Saint George International School

Enterprise

PORTFOLIO

By: JAF Racing Team



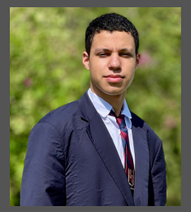


Driven by Innovation, United by Ambition

JAF RACING TEAM

JOURNEY • AMBITION • FOCUS

TEAM IDENTITY

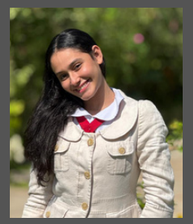












JESUS LOPEZ — Project Manager

Leads the team, coordinates the different areas, organizes the project timeline, and ensures that all team members work toward the same racing goals.

BRIAN LI — Racing Driver & Performance Engineer

Responsible for driving and testing the car. Analyzes the car's performance during races and helps identify areas for improvement.

DUNIEL CHOCK — Design & Engineering Lead

Develops the car's technical and aerodynamic design. Evaluates dimensions, structure, and performance to improve the overall efficiency of the vehicle.

RACHEL CARTER— Marketing & Communications Manager

Leads the team's marketing strategy, communication, and public presentation. Helps develop the team identity and promote JAF Racing Team.

NICOLE ZHONG — Team Support Coordinator

Supports the team's organization and race preparation. Helps coordinate materials, logistics, documentation, and other activities needed for the project.

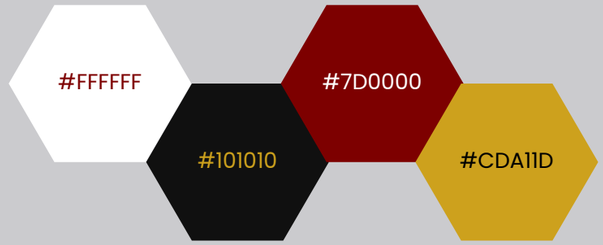
ANDREA DE CASTRO — Graphic & Visual Design Lead

Creates and manages the team's visual identity, including the logo, graphics, presentation materials, car decals, and overall visual style.

COLOURWAY

Our colourway represents the energy, speed, and competitive spirit of JAF Racing Team. We chose red for speed, power, and determination, gold for excellence and ambition, and black for a strong, professional contrast. Together, these colours create a bold and unified racing identity.

COLOR CODES



TEAM UNIFORM DESIGN

Our uniform uses the same red, gold, and black colourway to create a consistent team identity. We focused on the logo, colour combinations, and racing-inspired details to achieve a clean, professional, and unified look for all team members.



Figure 1. Uniform Concepts

LOGO EVOLUTION

This figure shows the evolution of our JAF Racing Team logo. We created two main versions, experimenting with the colour of the A and the three key words below the logo. In the first version, these elements were gold, while in the second version, they were changed to white for a cleaner and stronger contrast, while keeping the same racing identity.

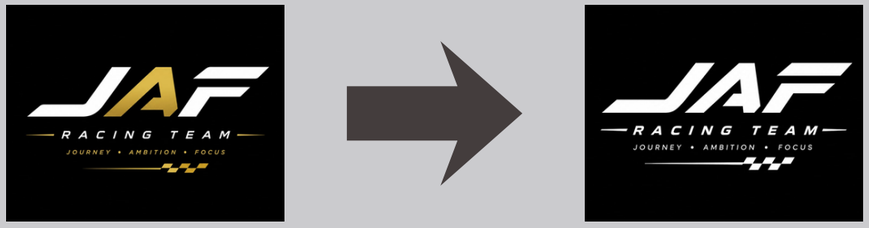


Figure 2. Logo Progression

MANAGEMENT & TEAMWORK

TEAM ORGANIZATION

From the first day, our team focused on creating a clear organization where every member had a specific role and responsibility. We started by identifying each person's strengths, skills, and interests, and then divided the different areas of the project accordingly. This allowed us to distribute the workload fairly and make sure that every important part of the project had someone responsible for it. At the same time, we understood that having separate roles did not mean working individually, so we made sure to support each other whenever a task required more than one person. We regularly checked our progress and adjusted responsibilities when necessary to keep the project moving forward.

Communication was one of the most important parts of our teamwork. We constantly shared ideas, discussed problems, and updated each other about the progress of our tasks. Before making important decisions, especially changes to the car or its design, we listened to everyone's opinions and considered different solutions before choosing the best option. If someone was having difficulty with a task, other members were willing to help or give suggestions instead of leaving the person to solve the problem alone. This created a more collaborative environment where everyone could contribute their ideas and feel involved in the project.

Our organization also helped us manage our time and avoid unnecessary delays. We divided larger objectives into smaller tasks and made sure that responsibilities were clear, so everyone knew what needed to be done and when it needed to be completed. We communicated when a task was finished, when something needed to be changed, or when another member's work depended on it. This was especially important because different areas of the project were connected, meaning that a delay in one task could affect the work of another member. By keeping each other informed, we were able to coordinate our work more effectively and maintain a steady workflow.

We also made decisions as a team rather than individually. When we had different opinions, we discussed the advantages and disadvantages of each idea and tried to reach an agreement that benefited the overall project. We learned that effective teamwork was not simply about dividing the work, but also about trusting each other, communicating clearly, and taking responsibility for our assigned areas. This organization allowed us to combine our individual strengths and work toward the same goal.

TEAM NAME

The name **JAF** comes from three words that are very important to us and are also part of our team's motto: **Journey**, **Ambition**, and **Focus**.

Journey represents our learning process, which began with learning how to use the software and continued through learning how to design the race car. **Ambition** is about how we want to reach the very top and compete with all our might. Last but not least, **Focus** is the discipline of always listening to one another and making sure every member works toward the same goal. Together, these words became our initials and guide every decision we make as a team.

TEAM GOAL

Our goal is to design, build, test, and improve a competitive **STEM** racing car while working effectively as a team. We aim to combine engineering, design, creativity, and teamwork to achieve the best possible performance.

EVALUATION OF MANAGEMENT AND TEAMWORK

Our management and teamwork were effective because we assigned responsibilities according to each member's strengths and maintained good communication throughout the project. Using tools such as **Trello**, **Autodesk Fusion 360**, **Canva**, **Word**, and **Creality Print** helped us organize our work and complete each stage efficiently.

We also made good use of the resources available to each team member, allowing us to divide tasks effectively and avoid unnecessary delays. Overall, our communication, organization, and teamwork helped us solve problems, improve our design, and stay focused on our goal.

MARKETING & DEVELOPMENT TOOLS

BRAND IDENTITY

Our marketing strategy focused on creating a strong and consistent identity for JAF Racing Team. We developed our logo, colourway, team uniform, and car graphics using red, gold, and black to make our team easily recognizable.

DIGITAL MEDIA

We used a combination of **Canva**, **Instagram**, and video editing tools such as **CapCut** to showcase our team, car, designs, and progress. These tools helped us create photos, videos, and promotional content to present JAF Racing Team and increase our visibility.

Figure 3. Marketing tools

EVALUATION OF MARKETING

Our marketing helped us build a recognizable team identity and present JAF Racing Team professionally. By combining social media, photography, posters, car stickers, and consistent visual design, we were able to communicate our team's personality and racing spirit. We made sure that our logo, colours, and overall style remained consistent across our different materials, creating a stronger connection between our physical car and digital presence. These efforts also helped us showcase the work behind our project, including our designs, progress, and teamwork, while making our team more engaging and memorable to our audience.

TEAM & CAR DESIGN

Our uniform and car stickers were designed to match our logo and colourway, creating a unified appearance. This helped connect our physical car with our digital and promotional materials.

Figure 4. Car Stickers Designs

AUDIENCE

We wanted our content to attract people interested in STEM, racing, engineering, and Formula 1. By sharing our designs, car development, and team progress, we could show both the technical and creative side of our project.

TOOLS & SOFTWARE

Throughout the project, we used different digital tools to support each stage of our work. **Trello** helped us organize our tasks, assign responsibilities, and monitor our progress. **Fusion 360** was used to design the car, develop its components, and make adjustments to improve the final design. **ANSYS** helped us analyze the car's design and evaluate its aerodynamic performance through simulations. **Word** was used to document our research, planning, and project information. Finally, **Creality Print** was used to prepare our designs for 3D printing, allowing us to manufacture components such as the spoiler and front wing. Together, these tools supported our workflow, from the initial design concepts to the physical car and its evaluation.

Figure 5. Team Tools and Software

VISUAL PROMOTION

We designed posters and promotional graphics to present our team in a professional and attractive way. The same visual identity was maintained across our posters, social media, uniform, and car stickers.



Figure 6. Instagram Post Series

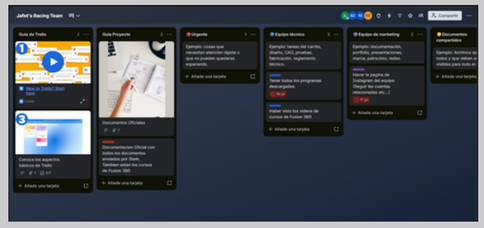


Figure 7. A screenshot of our Trello workspace, showing our team objectives and tasks organized by their progress, from completed to ongoing and not started.

PROJECT MANAGEMENT & PLANNING PROJECT TIMELINE

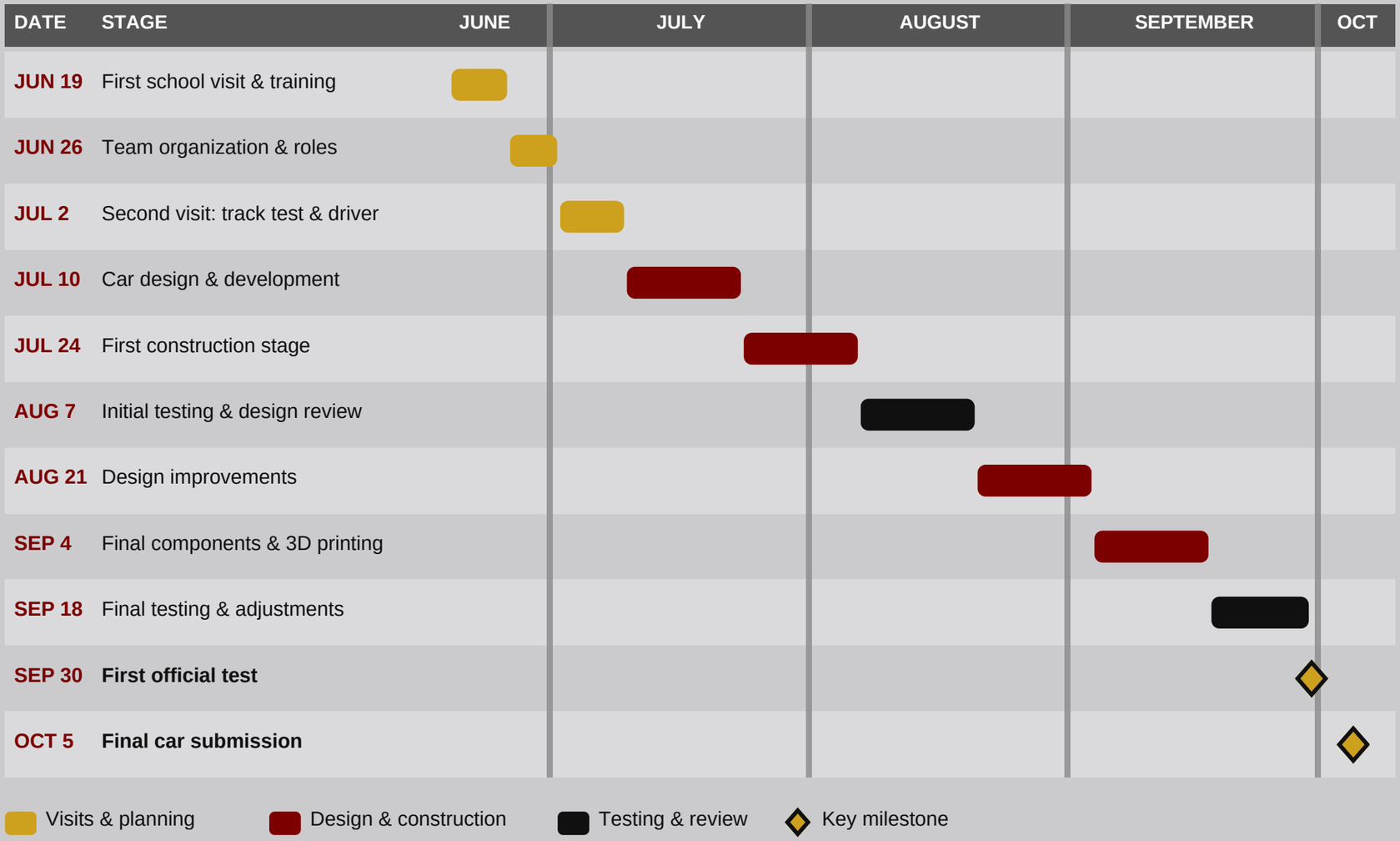


Figure 8. *JAF Racing Team project timeline. Each bar runs from the start of a stage to the start of the next one; the diamonds mark the first official test (September 30) and the final car submission (October 5).*

A Gantt chart is a project management tool that helps organize the different tasks of a project by showing what needs to be done, when it needs to be done, and how long each task will take. We used a Gantt chart to plan and divide the work of the JAF Racing Team throughout the entire project. Instead of completing tasks without a clear order, we organized our work into different stages, starting with the first school visit and team organization and continuing through the car design, construction, testing, improvements, marketing, and final submission.

The Gantt chart allowed us to clearly see the timeline and progression of our project. We divided larger objectives into smaller tasks and placed them according to the dates when they needed to be completed. This made it easier to understand which tasks were a priority and which ones depended on previous work being finished. It also helped us coordinate the different areas of the team because some tasks had to be completed before others could begin. For example, the initial design had to be developed before we could produce and test physical components, and the testing results had to be reviewed before making further improvements to the car.

We also used the Gantt chart to help with the division of responsibilities. Each member could identify the tasks they were responsible for and understand how their work contributed to the overall project. This helped us avoid confusion, distribute the workload more effectively, and make sure that important tasks were not forgotten. As the project progressed, we could also review what had already been completed and identify what still needed attention.

Another important benefit was that the Gantt chart helped us manage our time and stay organized. Since our project included several areas such as engineering, design, testing, documentation, marketing, and presentation, having everything organized in one timeline gave us a clearer view of the whole process. When changes or delays occurred, we could adjust our tasks and reorganize our priorities instead of losing track of the overall schedule.

Overall, the Gantt chart helped us turn our project into a structured and manageable process. It gave us a clear timeline, helped us divide responsibilities, coordinate our work, monitor our progress, and make sure that each stage of the JAF Racing Team project was completed before moving on to the next one. It was an important part of keeping the team organized and working toward the same final goal.

GRAPHIC DESIGN & VISUAL DEVELOPMENT

Graphic design is a way of expressing ideas and communicating information through visual elements. Throughout our project, we used graphic design to represent our thought processes, document the development of our ideas, and communicate the different stages of JAF Racing Team. It also helped us simplify complex concepts through visualization, making our designs easier to understand and evaluate.

For example, using Fusion 360, we created 3D models of our racing car and its components, allowing us to examine their shapes, proportions, and overall structure before moving forward with the design.

These models helped us visualize how the different components would fit together and communicate our ideas more clearly. In addition, we used ANSYS to support our design analysis through simulations, helping us evaluate aerodynamic behavior and better understand how our car's design could perform.

Digital visualization allowed us to examine our car from different perspectives and identify possible improvements during the design process.

Branding also plays an important role in giving our team a recognizable identity and communicating our personality, objectives, and competitive spirit.

We developed visual materials that connected the technical and creative sides of our project, including our presentations, promotional graphics, posters, and social media content. We used Canva to create and organize these materials, ensuring that our information was presented clearly and professionally.

We also used Instagram to showcase our progress and share content about our team, while CapCut helped us edit videos to make our digital content more engaging. Each of these tools served a different purpose, but together they helped us communicate our work and present JAF Racing Team to a wider audience.

EXHIBITION DISPLAY



Figure 9. Exhibition Display Designed for the JAF Racing Team Presentation.

PROMOTIONAL BUSINESS CARD



Figure 10. Promotional Business Card Designed to Showcase the JAF Racing Team and Its Instagram QR Code.

Another important part of our graphic design process was developing digital concepts for the physical car. Our car design was created in Fusion 360, while Creality Print allowed us to prepare selected components, such as the spoiler and front wing, for 3D printing. This allowed us to develop our designs digitally and prepare the components according to our project requirements. Throughout the process, we reviewed our designs and made adjustments when necessary, ensuring that our ideas continued to develop as we evaluated different possibilities. Overall, graphic design was not only about creating attractive visuals; it was also a way to communicate technical information, support decision-making, and connect our engineering work with the public presentation of our project. By combining 3D modelling, simulation, digital graphics, and visual communication, we were able to develop a more complete and professional representation of JAF Racing Team.

ACKNOWLEDGEMENTS

Being part of the **STEM Racing in Panama** project for the first time has been an unforgettable experience for our entire team. Throughout this journey, we had the opportunity to learn new skills, explore our creativity, apply our knowledge of science, technology, engineering, and mathematics, and discover what it truly means to work together toward a common goal.

From designing our car and developing our ideas to manufacturing components, testing our prototype, and improving our design, every stage brought new challenges, lessons, and moments of excitement. Although some parts of the process were challenging, overcoming them as a team made the experience even more rewarding. We are especially grateful to **Saint George International School of Panama** for giving us the opportunity to participate in this project and for supporting us throughout our journey.
We would also like to express our sincere appreciation to our teachers, mentors, and everyone who guided us, shared their knowledge, and encouraged us to keep moving forward.
A special thank-you goes to **STEM Racing** for creating a platform that allows students to experience engineering and innovation in such a practical, creative, and exciting way.

This project was something new for us, and it allowed us to step outside our comfort zone, develop confidence, strengthen our teamwork, and create memories that we will carry with us beyond the competition.
We feel proud of what we have accomplished as the **JAF Racing Team**, not only because of the car we designed, but also because of everything we learned and experienced along the way.
We finish this journey with gratitude, excitement, and a greater appreciation for engineering, collaboration, and innovation. This is more than just a racing project; it is a reminder that with ambition, dedication, and teamwork, we can turn our ideas into reality. Thank you for being part of our journey.